

String:

Properties:

```
In [ ]: String is immutable(can not change)
String is in ordered.
Duplicate items are allowed.
Single line string>>single quate and diuble quate
Multiline string>>triple singlke quates and triple double quates.
```

String Functions:

1)capitalize()

```
In [ ]: Fisrt alphabet of string capitalize.
```

```
In [4]: string1="data science"
string1.capitalize()
```

```
Out[4]: 'Data science'
```

2)upper()

```
change string into uppercase.
```

```
In [6]: string1="data science"
string1.upper()
```

```
Out[6]: 'DATA SCIENCE'
```

```
In [8]: str1="data science 1234"
str1.upper()
```

```
Out[8]: 'DATA SCIENCE 1234'
```

```
In [9]: str1="data science 1234 #%&*"
str1.upper()
```

```
Out[9]: 'DATA SCIENCE 1234 #%&*'
```

```
In [10]: str2="Data science is a multidisciplinary field that combines statistics, mach
str2.upper()
```

```
Out[10]: 'DATA SCIENCE IS A MULTIDISCIPLINARY FIELD THAT COMBINES STATISTICS, MACHINE
LEARNING, AND COMPUTER SCIENCE'
```

3)lower()

It will convert all letters in lowercase.

```
In [11]: string1="DATA SCIENCE AND PYTHON"  
string1.lower()
```

```
Out[11]: 'data science and python'
```

```
In [13]: str1="Data science is a Multidisciplinary field that combines statistics, mach  
str1.lower()
```

```
Out[13]: 'data science is a multidisciplinary field that combines statistics, machine  
learning, and computer science'
```

4)title()

First character of every word will be in uppercase.

```
In [14]: string1="virat kohali"  
string1.title()
```

```
Out[14]: 'Virat Kohali'
```

```
In [15]: str1="DATA SCIENCE"  
str1.title()
```

```
Out[15]: 'Data Science'
```

```
In [16]: str2="machine learning"  
str2.title()
```

```
Out[16]: 'Machine Learning'
```

```
In [17]: str3="Data science is a multidisciplinary field that combines statistics, mach  
str3.title()
```

```
Out[17]: 'Data Science Is A Multidisciplinary Field That Combines Statistics, Machine  
Learning, And Computer Science'
```

```
In [18]: Aadhar_name="virat kohli"
PAN_name="VIRAT KOHLI"

Aadhar_name=Aadhar_name.upper()
print("Adhar name:-",Aadhar_name)

PAN_name=PAN_name.upper()
print("PAN Name:-",PAN_name)

Aadhar_name==PAN_name
```

Adhar name:- VIRAT KOHLI
PAN Name:- VIRAT KOHLI

Out[18]: True

```
In [19]: Aadhar_name="virat kohli"
PAN_name="VIRAT KOHLI"

Aadhar_name=Aadhar_name.upper()
print("Adhar name:-",Aadhar_name)

PAN_name=PAN_name.upper()
print("PAN Name:-",PAN_name)

if Aadhar_name==PAN_name:
    print("Both strings are same")
else:
    print("Both strings are different")
```

Adhar name:- VIRAT KOHLI
PAN Name:- VIRAT KOHLI
Both strings are same

5)swapcase()

It inverts the case of characters(Alphabets)

```
In [20]: Adhar_name="virat kohli"
Adhar_name.swapcase()
```

Out[20]: 'VIRAT KOHLI'

```
In [21]: str1="Data Science"
str1.swapcase()
```

Out[21]: 'dATA sCIENCE'

6)lstrip()

It removes all leading white spaces from string.

```
In [22]: adhar_name="          virat kohli"
x=adhar_name.lstrip()
print(adhar_name)
print(x)
```

```
          virat kohli
virat kohli
```

7)rstrip()

It remove all trailing white spaces from string.

```
In [23]: string1="python and data science      "
string1.rstrip()
```

```
Out[23]: 'python and data science'
```

```
In [24]: string1="          python and data science      "
string1.rstrip()
```

```
Out[24]: '          python and data science'
```

8)strip()

It removes leading and trailing white spaces.

```
In [25]: str1="          python and data science      "
str1.strip()
```

```
Out[25]: 'python and data science'
```

9)replace()

Replace old string with new string.
Syntax:-replace("old_string","new_string")

```
In [26]: string="Python and data sceince"
string.replace("Python","Machine Learning")
```

```
Out[26]: 'Machine Learning and data sceince'
```

10)count()

```
In [ ]: count the occurrences of the substring in a string.  
Syntax:- string.count(substring,[start_index],[end_index])
```

```
In [27]: string="python and data science , python is used for deep learning , python is  
string.count("python")
```

Out[27]: 3

```
In [28]: string="python and data science , python is used for deep learning , python is  
string.count("is")
```

Out[28]: 2

```
In [29]: string="python and data science , python is used for deep learning , python is  
string.count("a")
```

Out[29]: 4

```
In [30]: string="python and data science , python is used for deep learning , python is  
string.count("e")
```

Out[30]: 7

```
In [31]: string="python and data science , python is used for deep learning , python is  
string[0:5]
```

Out[31]: 'pytho'

```
In [32]: string="python and data science , python is used for deep learning , python is  
string[0:40]
```

Out[32]: 'python and data science , python is used'

```
In [33]: string="python and data science , python is used for deep learning , python is  
string.count("python",0,40)
```

Out[33]: 2

```
In [36]: string="python and data science , Python is used for deep learning , Python is  
string.lower().count('python')  
#first string convert into lowercase then count the substring occurrences
```

Out[36]: 4

```
In [37]: string="python and data science , Python is used for deep learning , Python is  
string.count('python')
```

Out[37]: 2

11)index()

To findof the index of substring
Syntax:-string.index(substring,[start_index,end_index])

```
In [38]: string1="Machine Learning and deep learning"
string1.index("Machine")
```

Out[38]: 0

```
In [41]: string1="Machine Learning and deep learning"
string1.index("Machine")
```

Out[41]: 0

```
In [49]: string1="Machine Learning and deep learning"
print(string1.count("a"))
print(string1.index("a"))
```

4
1

```
In [48]: string1="Machine Learning and deep learning"
string1.index("a")
```

Out[48]: 1

```
In [47]: string1="Machine Learning and deep learning"
string1.index("and")
```

Out[47]: 17

12)find()

```
In [ ]: To find substring is present in string or not , return index
Syntax:- string.find(substring,[start_index],[end_index])
Return index
```

```
In [50]: string="Machine learning and deep learning"
string.index("learning")
```

Out[50]: 8

```
In [51]: string="Machine learning and deep learning"
string.find("learning")
```

Out[51]: 8

```
In [52]: string="Machine learning and deep learning"
         string.index("python")
```

ValueError Traceback (most recent call last)

Cell In[52], line 2

```
1 string="Machine learning and deep learning"
----> 2 string.index("python")
```

ValueError: substring not found

```
In [55]: string="Machine learning and deep learning"
         string.index("learning",10,20)
```

ValueError Traceback (most recent call last)

Cell In[55], line 2

```
1 string="Machine learning and deep learning"
----> 2 string.index("learning",10,20)
```

ValueError: substring not found

```
In [56]: string="Machine learning and deep learning"
         string.index("learning",10)
```

Out[56]: 26

13)split()

It is used to convert into list data type.
The split() function separate the given string by
the defined delimiter i.e space(' ')

```
In [62]: string="Machine Learning and Data science and deep learning"
         x=string.split()
         print(x)
         print(len(x))
```

```
['Machine', 'Learning', 'and', 'Data', 'science', 'and', 'deep', 'learning']
8
```

```
In [60]: string="Machine Learning and Data science and deep learning"
         string.split(" ")
```

Out[60]: ['Machine', 'Learning', 'and', 'Data', 'science', 'and', 'deep', 'learning']

```
In [65]: string="Machine Learning and Data science and deep learning"
x=string.split('and')
print(x)
len(x)
```

```
['Machine Learning ', ' Data science ', ' deep learning']
```

```
Out[65]: 3
```



```
In [66]: txt="""Data science is a multidisciplinary field that combines statistics,
machine learning, and computer science to extract actionable
insights from raw data. It is used across industries to clean,
analyze, and model large datasets, helping organizations predict
future trends, optimize operations, and make informed strategic decisions"""
txt.split()
```

```
Out[66]: ['Data',
'science',
'is',
'a',
'multidisciplinary',
'field',
'that',
'combines',
'statistics,',
'machine',
'learning,',
'and',
'computer',
'science',
'to',
'extract',
'actionable',
'insights',
'from',
'raw',
'data.',
'It',
'is',
'used',
'across',
'industries',
'to',
'clean,',
'analyze,',
'and',
'model',
'large',
'datasets,',
'helping',
'organizations',
'predict',
'future',
'trends,',
'optimize',
'operations,',
'and',
'make',
'informed',
'strategic',
'decisions']
```

```
In [68]: a="Gadient.Decent.png"
a.split('.')[0]
```

```
Out[68]: 'Gadient'
```

In []: